

TWO-WAY PROPERTY RATE DIGITAL ADMINISTRATION PLATFORM

Technical Architecture & Operational Specification Document

SYSTEM SCOPE & IMPLEMENTATION DIRECTIVES:

- Two-Way Interaction Model:** The platform establishes a synchronized digital bridge between Municipal Revenue Administration (Client/Assembly) and Property Rate Payers (Citizens).
- Consolidated In-App Settlement:** Replaces disconnected SMS hyperlinks with a unified, in-app digital demand notice and integrated payment gateway engine.
- Automated Assessment Rollout:** Standardizes annual batch billing, carried-forward arrears consolidation, and automated notification dispatch.

SECTION 1 — BASELINE DATA STRUCTURE & FIELD SPECIFICATION

Core Data Schema, Field Definitions, and Architectural Purpose

CATEGORY	FIELD IDENTIFIER	DATA TYPE & FORMAT	FUNCTIONAL DEFINITION & SYSTEM PURPOSE
B. Property & Account Identifiers	Account No.	Alphanumeric (Indexed)	Primary Account Head and unique municipal property identifier; acts as master database key for records, lookup, and payment reconciliation.
	Owner / Digital Address	Alphanumeric (GPS Format)	GhanaPost GPS digital address / Taxpayer ID; ties the property account to its physical location coordinates.
	Property Classification	Enumerated Category	Zoning and property classification (e.g., Residential Classes, Commercial) determining the applied rating factor.
D. Billing Timeline & Deadlines	Bill Year	Integer (YYYY)	Active assessment fiscal year for statutory municipal rating and billing cycles.
	Bill Date (Issued)	Date (DD-MMM-YYYY)	Official timestamp of demand notice generation and dispatch to the taxpayer.
	Settlement Deadline (Due Date)	Date (DD-MMM-YYYY)	Statutory cutoff date (e.g., June 30), after which unsettled accounts are flagged for administrative follow-up.
E. Valuation & Rate Computation	Rateable Value	Decimal / Currency	Certified property capital valuation assessed by rating authorities.
	Rate Imposed (Rate-in-the-Cedi)	Decimal Multiplier	Statutory rate multiplier applied against the Rateable Value to compute the current fee.
F. Financial Breakdown (Bill Table)	Previous Year Bill	Decimal / Currency	Total assessed fee billed during the preceding fiscal year.
	Amount Paid (Last Yr)	Decimal / Currency	Total payments credited against the previous year's bill.
	Arrears	Decimal / Currency	Cumulative outstanding debt carried forward from prior unpaid cycles.
	Current Fee	Decimal / Currency	Active period fee levied for the assessment year (Rateable Value × Rate Imposed).
	Total Amount Due	Decimal / Currency	Consolidated balance payable by the taxpayer (Arrears + Current Fee).

SECTION 2 — TWO-WAY SYSTEM ARCHITECTURE

Functional Modules, Integration Layers, and Bidirectional Workflows

MODULE A: TAXPAYER MOBILE INTERFACE (CITIZEN APPLICATION)

- 1. Telephone Authentication & Account Number Resolution:** Taxpayers authenticate using their mobile telephone number via SMS OTP. The telephone number serves strictly as an authentication and verification mechanism. Once verified against records present in the database, the system resolves and displays all property **Account Numbers (Account Heads)** registered under that contact number.
- 2. Native Digital Demand Notice Display:** Provides an authenticated, mathematically transparent demand notice interface displaying Account No., Owner/Digital Address, Classification, Rate Calculation Formula, and Itemized Ledger in real time.
- 3. Integrated In-App Settlement Engine:** A native payment engine embedded directly on the bill supporting modular settlement options:
 - **Full Balance Settlement:** Settle total consolidated outstanding liability.
 - **Arrears Settlement Only:** Settle carried-forward prior debt independently.
 - **Active Year Assessment Only:** Settle current period fee independently.
- 4. Multi-Channel Payment Gateway:** Direct integration with Mobile Money payment rails (MTN Mobile Money, Telecel Cash, AT Money) and Bank Debit/Credit cards.
- 5. Automated Proof of Payment Issuance:** Upon instant payment gateway confirmation, the system generates an official authenticated digital receipt stored permanently within the taxpayer's account ledger.

MODULE B: MUNICIPAL ADMINISTRATION & REVENUE AUTOMATION PORTAL

- 1. Automated Annual Batch Billing Engine:** Scheduled annual computation engine triggered during the opening fiscal window to compute assessments across all registered properties:

$$\text{Current Fee} = \text{Rateable Value} \times \text{Rate Imposed}$$
$$\text{Total Amount Due} = \text{Arrears} + \text{Current Fee}$$

- 2. Automated SMS Broadcast & Secure Token Dispatch:** Mass messaging pipeline that delivers concise statutory billing alerts to telephone numbers registered on each property **Account Number**, containing secure single-session token links that open the specific bill directly within the mobile application.
- 3. Revenue Analytics & Defaulter Tracking:** Administrative dashboard providing real-time collection metrics filtered by Property Classification and Billing Status. Accounts with unpaid balances past the statutory settlement deadline (June 30) are automatically flagged for administrative follow-up.

SECTION 3 — END-TO-END OPERATIONAL WORKFLOW

Lifecycle Stages, Execution Hierarchy, and Financial Ledger Principles

OPERATIONAL PHASE	SYSTEM ACTION & EXECUTION LOGIC	RESPONSIBLE PARTY	OUTPUT / DELIVERABLE
1. Annual Rollout	System executes batch computation; assesses current year fees and rolls uncollected previous balances into arrears.	Municipal Revenue Unit	Annual Assessment Roll
2. Notification Dispatch	Automated SMS engine broadcasts demand notices with encrypted deep-links to registered mobile telephone numbers tied to each property Account Number.	Automated Messaging Service	Direct Taxpayer Notification
3. Demand Notice Ingestion	Taxpayer opens secure link; reviews itemized assessment, valuation factor, and arrears balance.	Property Rate Payer	Authenticated Bill View
4. In-App Settlement	Taxpayer selects settlement amount (total amount due, arrears, or current fee) and authorizes transaction via payment gateway.	Payment Gateway / Payer	Real-Time Settlement Webhook
5. Ledger Reconciliation	System database updates property ledger, reduces balance, and issues official digital receipt.	Core Database Engine	Official Digital Receipt
6. Statutory Enforcement	Accounts with outstanding balances past statutory deadline (June 30) are flagged for administrative follow-up.	Revenue Enforcement Unit	Compliance & Follow-up List

FINANCIAL LEDGER & ACCOUNTING PRINCIPLES

- Priority Debt Allocation:** All incoming payments are applied strictly against the oldest debt first. Settlements reduce carried-forward **Arrears** before liquidating the **Current Fee**.
- Partial Settlement Accounting:** When partial payments are received, the system adjusts the arrears balance in real time while maintaining the active year fee structure until fully settled.
- Immutable Audit Logging:** Every billing transaction, SMS dispatch event, bill view, and payment attempt is logged with an immutable timestamp, operator ID, and network reference for municipal audit compliance.